

Assessment Specification – Portfolio Task 1

Module Details

Module Code	UFCE8B-30-3
Module Title	Data Science for Cyber Security
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Year	2024-25
Task	01
Total number of assessments for this module	01
Weighting	20%
Date issued to students	24/10/2025

Task Specification

You are a cyber security analyst building a data analysis pipeline to train and test a packet classifier. You have acquired a labelled network traffic dataset that has 47 network features for each packet (such as, source ip, source port, protocol etc.). The features are described in a CSV file titled 'features.csv'. The data has two class label columns as well; one that labels malicious packets with the attack type out of nine possible options (*namely, Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode and Worms.*) and another that labels the data as benign/malicious.

Before you use the data from the dataset to train a machine learning model, you are planning to run descriptive and exploratory analysis focussed at cleaning the data. Further focus of data cleaning is on identifying and fixing missing, incomplete and inaccurate data.

Dataset: task1.csv (Available on Blackboard.)

You must complete this portfolio task using Python programming language and Jupyter notebook.

Deliverable: Prepare a single Jupyter notebook.

Using the dataset provided you have to perform the following tasks,

	Task	Marks
1	Load the dataset, find out and print the total number of rows and columns in the dataset.	2
2	Run exploratory analysis on the dataset and print a summary of missing values across all columns. Fix the missing values using an interpolation method.	3

3	Investigate if the source and destination IP columns have any incomplete or inaccurate IPs. Create a bar chart to show the number of errors, missing and correct values.	6
4	'Stime' and 'Ltime' shows two timestamps related to each row. Their description is in the features.csv file. Using your knowledge of cyber security and network traffic investigate if the two columns have missing or inaccurate values. If you find any missing or inaccurate values. Provide a fix for those issues and print the evidence on screen.	4
5	The dataset contains malicious traffic related to nine different attacks alongwith benign traffic. Find the percentage of malicious and benign packets in the sample. <i>Hint: Malicious traffic data points are labelled with the 'attack type' in one of the columns.</i>	2
6	In order to evaluate the distribution/share of each attack type, visualise the number of packets per attack type in an appropriate chart.	3
	Total	20
